

# **Lab Task 1**

## **CSC-201 Artificial Intelligence**



Session: 2023 – 2027

**Submitted by:**

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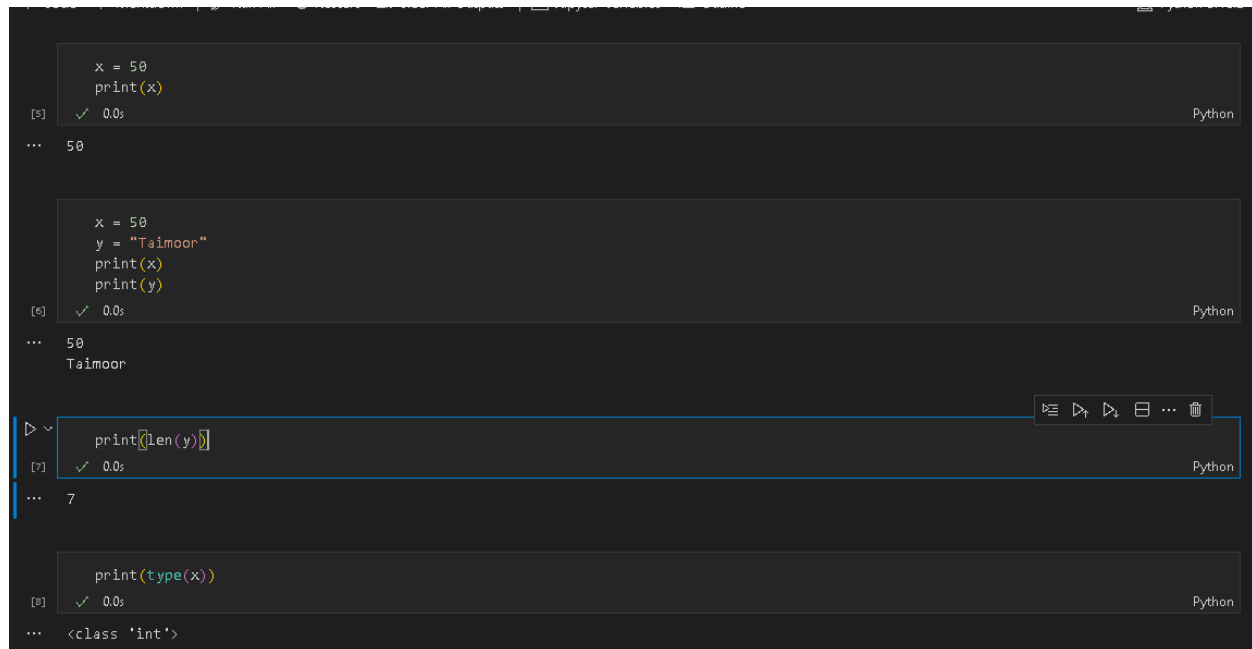
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**Submitted to:**

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# Jupyter notebook



```
x = 50
print(x)
```

50

```
x = 50
y = "Taimoor"
print(x)
print(y)
```

50  
Taimoor

```
print(len(y))
```

7

```
print(type(x))
```

<class 'int'>

## **Description:**

The file is saved with extension .ipynb.

### **First Block:**

Assigns `x = 50` and prints `x`. Output is 50.

### **Second Block:**

Assigns `x = 50` and `y = "Taimoor"`. Prints both variables. Outputs:

50

Taimoor.

### **Third Block:**

Prints the length of `y` (the string "Taimoor"). Output: 7.

### **Fourth Block:**

Prints the type of `x`. Output: `<class 'int'>`.

Each block demonstrates basic Python operations: assignment, printing, string length, and type checking.

# Program 1

```
1  a = 678
2  b = 888
3  print(a + b)
4
5  x2x = 123
6  print(x2x)
7
```

## Output:

```
PS E:\Artificial Intelligence> pyth
1566
123
PS E:\Artificial Intelligence>
```

## Description:

### **First Part:**

Assigns 678 to a and 888 to b.

Prints the sum of a and b ( $678 + 888 = 1566$ ).

### **Second Part:**

Assigns 123 to x2x.

Prints the value of x2x.

### **Key Notes:**

Variables can hold numbers and follow naming conventions (e.g., x2x is valid).

Simple addition and variable printing are demonstrated.

## Program 2

```
capitalize.py > ...  
1  variable = 'a'  
2  print(variable.capitalize())  
3  |
```

### Output:

```
PS E:\Artificial Intelligence>  
A  
PS E:\Artificial Intelligence>
```

### Description:

This code snippet performs the following actions:

**variable = 'a':**

This line creates a variable named variable and assigns it the string value 'a'.

**print(variable.capitalize()):**

This line uses the capitalize() method on the variable. The capitalize() method converts the first character of the string to uppercase and leaves the rest of the string in lowercase.

Since the value of variable is 'a', the capitalize() method will convert it to 'A'.

The print() function then outputs the result, so the character 'A' will be printed to the console.

## Program 3

casting.py > ...

```
1  # CASTING
2
3  # integer to string
4  num = 123
5  num_str = str(num)
6  print(num_str) # Output will be '123'
7  print(type(num_str)) # Output will be <class 'str'>
8
9  # string to integer
10 num_str = '456'
11 num = int(num_str)
12 print(num) # Output will be 456
13 print(type(num)) # Output will be <class 'int'>
14
15 # string to float
16 float_str = '3.14'
17 pi = float(float_str)
18 print(pi) # Output will be 3.14
19 print(type(pi)) # Output will be <class 'float'>
20
21 # integer to float
22 num = 10
23 num_float = float(num)
24 print(num_float) # Output will be 10.0
25 print(type(num_float)) # Output will be <class 'float'>
26
```

```

27 # float to integer
28 pi = 3.14
29 pi_int = int(pi)
30 print(pi_int) # Output will be 3
31 print(type(pi_int)) # Output will be <class 'int'>
32
33

```

#### **Output:**

```

PS E:\Artificial Intelligence> python -u
123
<class 'str'>
456
<class 'int'>
3.14
<class 'float'>
10.0
<class 'float'>
3
<class 'int'>
PS E:\Artificial Intelligence> █

```

#### **Description:**

This code demonstrates type casting in Python, which is the process of converting a variable from one data type to another. It covers:

Integer to String: Using `str()` to convert an integer to a string.

String to Integer: Using `int()` to convert a string to an integer.

String to Float: Using `float()` to convert a string to a float.

Integer to Float: Using `float()` to convert an integer to a float.

Float to Integer: Using `int()` to convert a float to an integer, truncating the decimal part.

The code shows the results and data types after each conversion.

# Program 4

```
concatenation.py > ...  
1  # CONCATENATION  
2  a = 'abc'  
3  b = 'def'  
4  print(a + b)  
5  |
```

## Output:

```
PS E:\Artificial Intelligence> python -u  
abcdef  
PS E:\Artificial Intelligence> |
```

## Description:

This code demonstrates string concatenation in Python.

### **Breakdown:**

a = 'abc': A variable a is assigned the string value 'abc'.

b = 'def': A variable b is assigned the string value 'def'.

print(a + b): The + operator is used to concatenate (combine) the two strings a and b. The result of 'abc' + 'def' is 'abcdef'.

print(): The concatenated string 'abcdef' is printed.

## Program 5

```
curlyBrackets.py > ...
1  person = {
2      'name': 'Alice',
3      'age': 30,
4      'city': 'Wonderland'
5  }
6
7  print(person['name']) # Output will be 'Alice'
8  print(person) # Output will be {'name': 'Alice', 'age': 30, 'city': 'Wonderland'}
9
```

### Output:

```
PS E:\Artificial Intelligence> python -u "e:\Artificial Intelligence\curlyBrackets.py"
Alice
{'name': 'Alice', 'age': 30, 'city': 'Wonderland'}
PS E:\Artificial Intelligence>
```

### Description:

This code demonstrates working with a Python dictionary:

**person = {'name': 'Alice', 'age': 30, 'city': 'Wonderland'}:** A dictionary with key-value pairs.

**print(person['name']):** Prints 'Alice', the value associated with the key 'name'.

**print(person):** Prints the entire dictionary: {'name': 'Alice', 'age': 30, 'city': 'Wonderland'}.

## Program 6

```
AI_Lab_day_1 > datatypes.py > ...
1  u = 5 # integer
2  v = 5.0 # float
3  w = 'Taimoor' # string (using single quotation)
4  x = "Taimoor" # string (using double quotation)
5  y = True # boolean
6
```

### Description:



This code demonstrates the assignment of different data types in Python:

`u = 5`: An integer value is assigned to the variable `u`.

`v = 5.0`: A float value is assigned to the variable `v`.

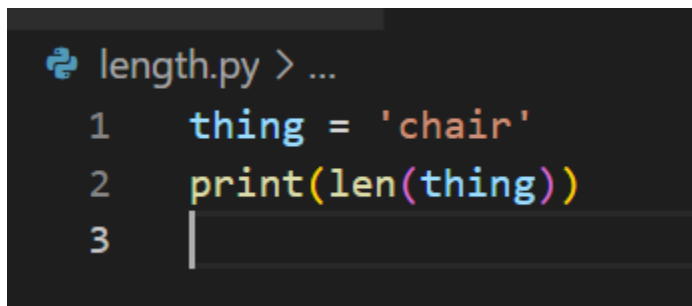
`w = 'Taimoor'`: A string is assigned to the variable `w` using single quotation marks.

`x = "Taimoor"`: A string is assigned to the variable `x` using double quotation marks.

`y = True`: A boolean value `True` is assigned to the variable `y`.

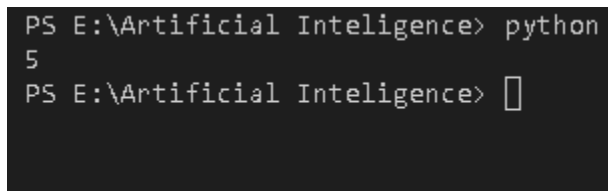
These variables represent different data types: integer, float, string, and boolean.

## Program 7



```
length.py > ...  
1  thing = 'chair'  
2  print(len(thing))  
3  |
```

### Output:



```
PS E:\Artificial Intelligence> python  
5  
PS E:\Artificial Intelligence> 
```

### Description:

This code demonstrates the use of the `len()` function to find the length of a string.

### **Breakdown:**

**`thing = 'chair'`**: A string `'chair'` is assigned to the variable `thing`.

**`print(len(thing))`**: The `len()` function is used to calculate the length of the string `'chair'`, which is 5 characters long. The result is printed.

## Program 8

```
lab_08_1 > multi_lower.py > ...  
1 variable = 't'  
2 print(variable.islower())  
3
```

### Output:

```
PS E:\Artificial Intelligence> python -  
True  
PS E:\Artificial Intelligence> []
```

### Description:

This code checks if the string variable consists of all lowercase letters using the `islower()` method.

### Breakdown:

**variable = 't':** A string 't' is assigned to the variable variable.

**print(variable.islower()):** The `islower()` method checks if all characters in the string variable are in lowercase. Since 't' is a lowercase letter, the method returns True.

## Program 9

```
multi_assigning.py > ...  
1 e,f,g = 1,2,3  
2 print(e,f,g)  
3
```

### Output:

```
PS E:\Artificial Intelligence> python  
1 2 3  
PS E:\Artificial Intelligence> []
```

### Description:

This code demonstrates multiple variable assignment and printing their values.

### **Breakdown:**

**e, f, g = 1, 2, 3:** The values 1, 2, and 3 are assigned to the variables e, f, and g, respectively.

**print(e, f, g):** The print() function is used to output the values of e, f, and g on the same line, separated by spaces.

## Program 10

```
parentheses.py > ...
1  def greet(name):
2      return f"Hello, {name}!"
3
4  print(greet("Alice")) # Output will be 'Hello, Alice!'
```

### Output:

```
PS E:\Artificial Intelligence> python -u "e:\Arti
Hello, Alice!
PS E:\Artificial Intelligence> □
```

### Description:

This code defines a function and uses it to greet a person by their name.

### **Breakdown:**

**def greet(name):** Defines a function greet that takes one parameter name.

**return f"Hello, {name}!":** Inside the function, it returns a formatted string that greets the person by their name using an f-string.

**print(greet("Alice")):** The function is called with the argument "Alice", so it returns the greeting "Hello, Alice!" and prints it.

# Program 11

```
1  # Slicing
2  name = 'Taimoor'
3  print(name[0:3])
4  print(name[3:])
5  print(name[:3])
```

## Output:

```
PS E:\Artificial Intelligence> python
Tai
moor
Tai
PS E:\Artificial Intelligence> 
```

## Description:

This code demonstrates string slicing in Python, which allows you to extract parts of a string.

### **Breakdown:**

**name = 'Taimoor':** A string 'Taimoor' is assigned to the variable name.

#### **print(name[0:3]):**

This slices the string starting from index 0 and up to (but not including) index 3.

It extracts the substring 'Tai'.

#### **print(name[3:]):**

This slices the string starting from index 3 and includes everything to the end.

It extracts the substring 'moor'.

#### **print(name[:3]):**

This slices the string from the beginning (index 0) up to (but not including) index 3.

It extracts the substring 'Tai' (same as the first slice).

# Program 12

```
squareBrackets.py > [?] fruits
1  fruits = ['apple', 'banana', 'cherry']
2
3  print(fruits[0]) # Output will be 'apple'
4  print(fruits)   # Output will be ['apple', 'banana', 'cherry']
5
```

## Output:

```
PS E:\Artificial Intelligence> python
apple
['apple', 'banana', 'cherry']
PS E:\Artificial Intelligence> []
```

## Description:

This code demonstrates how to work with a list in Python.

### **Breakdown:**

**fruits = ['apple', 'banana', 'cherry']:**

A list named fruits is created with three elements: 'apple', 'banana', and 'cherry'.

**print(fruits[0]):**

This accesses the element at index 0 in the fruits list, which is 'apple'.

**print(fruits):**

This prints the entire fruits list: ['apple', 'banana', 'cherry'].